

The Sun is a Deadly Laser: Solar Flare Prediction

Pradeepto Pal, Arush A S, and Abhinav C

Department of Computer Science and Engineering

SRM Institute of Science and Technology, Ghaziabad, Uttar Pradesh, India

Emails: pradeepto.home@gmail.com, arush.anand.singh.3121@gmail.com,
abhinavchauhan985@gmail.com

Abstract—Solar-flare forecasting is a rare-event prediction problem in which the most consequential events are also among the least frequent. This work presents a reproducible benchmark of five machine-learning approaches for predicting solar-flare activity from Solar Dynamics Observatory (SDO)/Helioseismic and Magnetic Imager (HMI) active-region observations. The benchmark uses the reduced-resolution active-region magnetogram dataset released by Boucheron *et al.*, containing 950,047 224×224 magnetogram images from 1,570 active regions with predefined active-region-grouped training, validation, and test partitions. Two complementary task formulations are evaluated. First, a CNN-LSTM model consumes ten-frame image sequences sampled at the native twelve-minute cadence and predicts one of three flare classes, C, M, or X. Second, four tabular classifiers—XGBoost, logistic regression, HistGradientBoosting, and Extra Trees—use the same 29 engineered magnetic-complexity features to detect flare thresholds of $\geq C$ and $\geq M$. On the held-out test set, the CNN-LSTM obtains 0.825829 accuracy and 0.444940 macro F1, while its X-class F1 is 0.0000 on only 34 X-class test examples. Among the tabular models, XGBoost gives the strongest ROC-AUC for both binary targets (0.890780 for $\geq C$ and 0.939496 for $\geq M$) and the highest reported TSS for the $\geq M$ task (0.757143). The $\geq M$ result also has 1.000000 recall but only 0.333333 precision, illustrating the importance of false-alarm behavior. Because the CNN-LSTM and tabular models solve different prediction tasks and operate on different representations, the results are interpreted as a complementary benchmark rather than a direct leaderboard. The study also isolates an available NASA SHARP-derived CSV as a future external-validation resource rather than mixing it into the current experiments.

Index Terms—Solar flare forecasting, solar active regions, SDO/HMI, magnetograms, CNN-LSTM, XGBoost, magnetic complexity features, rare-event classification, space weather

I. INTRODUCTION

Solar flares are abrupt releases of magnetic energy in the solar atmosphere. Their electromagnetic radiation can produce ionospheric disturbances and radio blackouts, while strong eruptive events and associated solar energetic particles can

affect spacecraft systems, navigation, communication, and other technological infrastructure [1], [2]. The operational value of flare forecasting therefore extends beyond academic understanding of solar activity: useful forecasts can provide lead time for mitigation actions in satellite operations, communications, navigation, and other space-weather-sensitive systems.

The flare-forecasting problem is difficult for two related reasons. First, the physical processes leading to magnetic energy storage and release are nonlinear and only partially observed at the photosphere. Second, the event distribution is strongly imbalanced. C-class or weaker periods are much more common than M- and X-class events, so a model can achieve high nominal accuracy while providing little practical forecasting skill. This issue has been emphasized repeatedly in the space-weather literature, motivating the use of metrics such as the True Skill Statistic (TSS) and Heidke Skill Score (HSS) in addition to conventional classification measures [11], [12].

The Solar Dynamics Observatory (SDO) provides a particularly useful observational basis for data-driven forecasting. Its Helioseismic and Magnetic Imager (HMI) continuously measures the photospheric magnetic field and associated active-region structure at high cadence [3]. Machine-learning studies have consequently explored both physically engineered active-region parameters and direct image representations. Classical approaches have demonstrated that magnetic gradients, neutral-line complexity, flux, and related quantities contain predictive information [4]–[6]. Deep-learning approaches have also shown that convolutional networks can learn discriminative spatial patterns from magnetograms [7], while recurrent models can exploit temporal evolution in active-region observations [9], [10], [13].

This paper develops a benchmark intended to make these

two modeling traditions comparable at the level of experimental protocol rather than by forcing them into an identical task. The image model is a CNN-LSTM that receives a short sequence of ten magnetogram frames and predicts C/M/X classes. The tabular branch evaluates four classifiers on 29 engineered magnetic-complexity features for two binary event thresholds, $\geq C$ and $\geq M$. The benchmark uses predefined active-region-grouped splits, selects operating thresholds on validation data, and reports held-out test metrics from the implemented training pipeline.

The main contributions are therefore: (1) a clearly separated image-sequence versus tabular benchmark on a public SDO/HMI dataset; (2) explicit reporting of rare-event metrics rather than relying on accuracy alone; (3) per-class analysis of the multiclass CNN-LSTM, including its failure on the rare X class; and (4) a validity discussion that distinguishes active-region grouping from true chronological forecasting and keeps a separate NASA/SHARP-derived CSV outside the reported training data.

II. RELATED WORK

A. Physics-based and feature-engineered forecasting

Solar-flare forecasting has a long history of relating flare occurrence to the magnetic complexity of active regions. Al-Ghraibah *et al.* studied magnetic gradients, neutral-line properties, flux, flux evolution, and wavelet descriptors and showed that photospheric complexity contains useful predictive information, while also emphasizing that these quantities are proxies rather than complete measurements of the coronal energy-release process [4]. Bobra and Couvidat subsequently demonstrated effective machine-learning flare prediction from SDO/HMI vector magnetic-field parameters, establishing a strong feature-based reference point for the field [5]. Florios *et al.* similarly evaluated machine-learning models using magnetogram-derived predictors for $> M1$ and $> C1$ forecasting within 24 h [6].

The present tabular branch is related to this feature-engineered tradition, but it uses a distinct set of 29 magnetic-complexity descriptors supplied with the benchmark dataset. These features summarize magnetic gradients, neutral-line geometry, multiscale wavelet structure, and flux. They are not the same as the SHARP metadata fields used in some operational systems, and this distinction is important when interpreting the results.

B. Deep learning and temporal modeling

CNN-based methods reduce reliance on manually designed descriptors by learning spatial representations directly from magnetograms. Huang *et al.* demonstrated deep-learning-based flare forecasting from line-of-sight magnetograms, while later studies considered richer image processing and direct active-region representations [7], [15]. Sequence modeling extends this idea by allowing the predictor to use the evolution of an active region. Liu *et al.* used LSTMs over sequences of magnetic and flare-history features, showing the usefulness of temporal information for 24-h flare prediction [9]. Chen *et al.* explicitly investigated time series of SDO/HMI observations and SHARP parameters and used deep learning to capture both spatial and temporal information [10].

Sun *et al.* studied CNN and LSTM models over two solar cycles and reported that temporal modeling could substantially improve TSS under some training configurations [13]. Zheng *et al.* later evaluated several deep architectures for multiclass forecasting, including bidirectional LSTM models with attention, and emphasized the value of evaluating both categorical skill and probabilistic reliability [14]. These studies motivate the present CNN-LSTM formulation, while also illustrating that performance is highly dependent on task definition, data partitioning, and the exact forecast target.

More recent work is shifting toward attention and transformer architectures. Alatoom and Nikolaou reported a 2026 multi-wavelength transformer approach for active-region-level forecasting using short sequences of SDO observations, while Riggi *et al.* evaluated foundational transformer models across image, video, and time-series modalities for solar-flare forecasting [21], [22]. These results indicate that temporal representation learning remains an active research direction; however, they do not invalidate CNN-LSTM or tabular models as useful controlled baselines, particularly when reproducibility and computational simplicity are priorities.

C. Forecast verification and rare-event imbalance

Bloomfield *et al.* argued for reliable benchmarking using forecast-verification measures and recommended TSS as a particularly useful statistic for solar-flare forecasting [11]. Camporeale's review similarly identifies class imbalance, evaluation methodology, uncertainty, and the combination of physical and data-driven information as central challenges for machine learning in space weather [12]. Accordingly, this study reports ROC-AUC and PR-AUC for ranking performance, together

TABLE I
PREDEFINED DATASET PARTITIONS.

Partition	Active Regions	Images	Fraction
Training	1,256	759,357	80%
Validation	157	95,933	10%
Test	157	94,757	10%
Total	1,570	950,047	100%

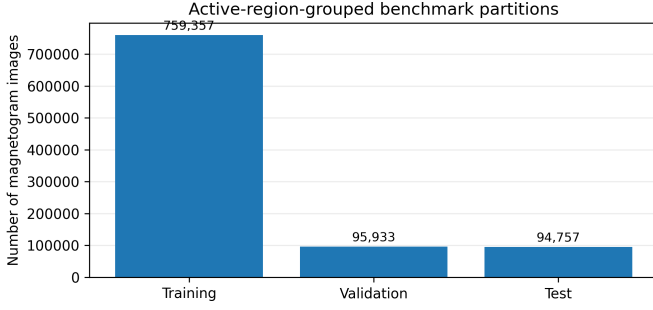


Fig. 1. Number of magnetogram images in the predefined active-region-grouped training, validation, and test partitions.

with precision, recall, F1, FAR, CSI, HSS, and TSS wherever these values are available from the saved experiments.

III. DATASET AND PROBLEM FORMULATION

A. Dataset provenance

The primary benchmark is the reduced-resolution active-region magnetogram dataset of Boucheron *et al.*, distributed through Dryad and described in *Scientific Data* [19]. The dataset was assembled from NOAA Space Weather Prediction Center active-region records, SDO/HMI magnetograms retrieved through the Joint Science Operation Center (JSOC), and GOES-based event reports. Magnetograms were centered on NOAA active regions and acquired at a native cadence of 720 s. The reduced-resolution release contains 950,047 224×224 pixel images spanning 1,570 active regions [20].

Unlike a random image-level split, the supplied partitions are grouped by active region. This prevents images from the same active region from appearing simultaneously in training and evaluation, reducing one important source of leakage. The source dataset deliberately adopts an approximately 80/10/10 active-region partition for this reason [19].

An important validity detail is that these partitions are not purely chronological. Timestamp ranges overlap across the three splits, so the experiments should be interpreted as active-region-grouped benchmark evaluation rather than evidence of future-time or cross-solar-cycle generalization.

B. Image labels and multiclass task

The supplied label file associates each magnetogram filename with a GOES flare class or with “0” for a non-flaring example. The present CNN-LSTM experiment reports a three-class problem over C, M, and X. The saved evaluation therefore does not represent the binary non-flare-versus-flare task used by the feature-based models.

For the image branch, ten frames are grouped chronologically within an active region. Consecutive accepted frames follow the native twelve-minute cadence, and candidate windows with missing or irregular timestamps are rejected. Each sequence receives the flare class associated with the forecast horizon following its final frame, yielding the three-way C/M/X prediction task implemented by the reported CNN-LSTM run.

C. Binary threshold tasks

The tabular experiments use two binary targets derived from the flare-class prefix. For a sample with label y , the targets are

$$y_{\geq C} = \begin{cases} 1, & y \in \{C, M, X\}, \\ 0, & y = 0. \end{cases} \quad (1)$$

and

$$y_{\geq M} = \begin{cases} 1, & y \in \{M, X\}, \\ 0, & y \in \{0, C\}. \end{cases} \quad (2)$$

These definitions are shared by all tabular experiments and are deliberately kept separate from the CNN-LSTM multiclass task.

D. Mathematical formulation of the learning problem

For the image branch, let a training example be a temporally ordered sequence $\mathcal{X} = \{x_1, x_2, \dots, x_T\}$, where $T = 10$ and each $x_t \in \mathbb{R}^{224 \times 224}$ is a normalized magnetogram. The CNN encoder maps each image to a latent vector

$$z_t = f_{\theta}(x_t), \quad z_t \in \mathbb{R}^d. \quad (3)$$

A generic convolutional layer can be expressed as

$$Z^{(\ell+1)} = \phi \left(W^{(\ell)} * Z^{(\ell)} + b^{(\ell)} \right), \quad (4)$$

where $*$ denotes convolution and ϕ is a nonlinear activation function.

The sequence model produces hidden states h_t and cell states c_t . For an input representation z_t , the LSTM gates can be written as

$$i_t = \sigma(W_i z_t + U_i h_{t-1} + b_i), \quad (5)$$

$$f_t = \sigma(W_f z_t + U_f h_{t-1} + b_f), \quad (6)$$

$$o_t = \sigma(W_o z_t + U_o h_{t-1} + b_o), \quad (7)$$

$$\tilde{c}_t = \tanh(W_c z_t + U_c h_{t-1} + b_c), \quad (8)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (9)$$

$$h_t = o_t \odot \tanh(c_t), \quad (10)$$

where $\sigma(\cdot)$ is the logistic sigmoid and $\odot$ is element-wise multiplication [18].

The multiclass probability for class k is computed with a softmax layer:

$$p_k = \frac{\exp(w_k^\top h_T + b_k)}{\sum_{j=1}^K \exp(w_j^\top h_T + b_j)}, \quad K = 3. \quad (11)$$

The corresponding cross-entropy objective is

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K y_{nk} \log p_{nk}. \quad (12)$$

For the tabular branch, each observation is represented by $x \in \mathbb{R}^{29}$. Logistic regression estimates the probability

$$P(y = 1 | x) = \sigma(\beta_0 + \beta^\top x) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top x)}}. \quad (13)$$

Its binary cross-entropy loss is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (14)$$

Training-set median imputation and standardization are applied before fitting. For a feature x_j , standardization is

$$x'_j = \frac{x_j - \mu_j}{\sigma_j}, \quad (15)$$

where μ_j and σ_j are estimated using the training partition only.

XGBoost represents the prediction as an additive tree ensemble

$$\hat{y}_i = \sum_{m=1}^M f_m(x_i), \quad f_m \in \mathcal{F}, \quad (16)$$

TABLE II
GROUPS OF THE 29 ENGINEERED MAGNETIC-COMPLEXITY FEATURES.

Feature group	Count
Magnetic-gradient statistics	7
Neutral-line geometry/statistics	13
Wavelet energy features	5
Flux features	4
Total	29

with regularized boosting objective

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(f_m), \quad (17)$$

where l is the training loss and Ω controls tree complexity [16].

E. Magnetic-complexity features

The feature CSV contains 29 engineered descriptors extracted from the reduced-resolution magnetograms. Following the published dataset description, they comprise seven gradient features, thirteen neutral-line features, five wavelet features, and four flux features [4], [19]. Representative quantities include gradient statistics, neutral-line length and curvature statistics, wavelet energy at multiple scales, and total signed and unsigned flux.

The repository also contains a separate SHARP/NASA-related CSV, `sharp_mx-nf-all_2010-01-01_2024-04-30.csv`. It is not mixed into the reported benchmark because its schema and provenance require a separate feature extraction, label-alignment, and validation pipeline. It is therefore treated as a future external-validation resource rather than as a hidden source of training information.

IV. METHODS

A. Benchmark overview

The experimental design can be summarized as two parallel branches:

B. CNN-LSTM image model

The image model maps each frame to a learned latent representation and then processes the resulting sequence with an LSTM. Its spatial feature extractor is represented by $f_\theta(\cdot)$ and its recurrent representation by (h_t, c_t) . The final hidden

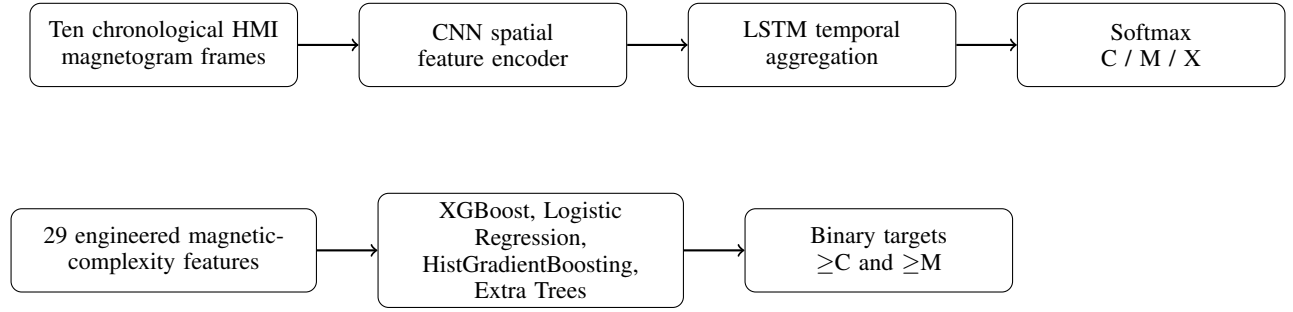


Fig. 2. Experimental architecture used for the benchmark. The CNN-LSTM branch operates on image sequences, whereas the four tabular models operate on the engineered 29-feature representation.

state is converted into a three-class probability distribution by the softmax layer described in the preceding section.

The LSTM formulation is appropriate for temporal dependencies because its gated memory mechanism was designed to preserve information over extended sequences [18]. In the present benchmark, the ten-frame input provides approximately two hours of temporal context from the first to the tenth observation, subject to sequence-validity checks.

The supplied evaluation artifact reports overall accuracy, macro F1, macro recall/POD, and macro forecasting skill scores, together with per-class F1. Architectural widths and other low-level details not present in the experiment record are not inferred.

C. XGBoost

XGBoost is a gradient-boosted decision-tree system that builds an additive ensemble of trees using regularized boosting objectives [16]. The present implementation uses the 29 tabular features with histogram-based training, learning rate 0.05, maximum tree depth 4, 500 estimators, subsample 0.8, column subsample 0.8, and random seed 42. Native missing-value handling is enabled, and early stopping is used where supported.

D. Logistic regression

Logistic regression provides a linear reference model for the same 29 features. The preprocessing pipeline performs median imputation using training data only, followed by standardization. This baseline separates the contribution of the feature representation from the additional nonlinear capacity of tree ensembles.

E. HistGradientBoosting

HistGradientBoosting is an independent histogram-based gradient-boosting implementation from scikit-learn rather than

a second implementation of XGBoost. The reported configuration uses 300 maximum iterations, learning rate 0.08, 31 maximum leaf nodes, L2 regularization of 1.0, and early stopping.

F. Extra Trees

Extra Trees forms an ensemble of highly randomized decision trees by randomizing feature selection and split-point selection [17]. The reported configuration uses 160 trees, square-root feature sampling, minimum leaf size 2, balanced class weights, all available CPU workers, and random seed 42.

V. EXPERIMENTAL PROTOCOL

A. Leakage control and threshold selection

The tabular loader validates the predefined training, validation, and test manifests and enforces active-region separation. The validation set is used for operating-point selection. For the reported tree-model results, candidate thresholds from 0.1 to 0.9 are evaluated and the threshold with maximum validation TSS is selected. That threshold is then applied once to the untouched test set. The test set is therefore not used to tune the operating threshold.

This protocol reduces leakage risk but does not turn the experiment into chronological forecasting. Since the dataset partitions overlap in time, the correct interpretation is grouped benchmark performance on unseen active regions.

B. Evaluation metrics

Let TP, TN, FP, and FN denote the entries of the binary confusion matrix. Probability of Detection (POD), equivalent to recall, is

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (18)$$

False Alarm Ratio (FAR) is

TABLE III
SUMMARY OF MODEL INPUTS AND PRINCIPAL REPORTED CONFIGURATIONS.

Model	Input	Task	Principal configuration
CNN-LSTM	10-frame HMI image sequence	Multiclass C/M/X	Convolutional encoder followed by LSTM temporal aggregation; ten chronological frames per sample.
XGBoost	29 engineered features	Binary $\geq C / \geq M$	Learning rate 0.05; depth 4; 500 estimators; subsample 0.8; column subsample 0.8; seed 42; native missing-value handling.
Logistic regression	29 engineered features	Binary $\geq C / \geq M$	Training-only median imputation followed by standardization; linear probability model.
HistGradientBoosting	29 engineered features	Binary $\geq C / \geq M$	300 iterations; learning rate 0.08; 31 maximum leaf nodes; L2 regularization 1.0; early stopping.
Extra Trees	29 engineered features	Binary $\geq C / \geq M$	160 trees; square-root feature sampling; minimum leaf size 2; balanced class weights; seed 42.

$$\text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}}. \quad (19)$$

The Critical Success Index (CSI) is

$$\text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \quad (20)$$

The F1 score is

$$F_1 = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}. \quad (21)$$

The True Skill Statistic is

$$\text{TSS} = \frac{\text{TP}}{\text{TP} + \text{FN}} - \frac{\text{FP}}{\text{FP} + \text{TN}}. \quad (22)$$

The Heidke Skill Score is

$$\text{HSS} = \frac{2(\text{TP TN} - \text{FP FN})}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})}. \quad (23)$$

These definitions follow established flare-forecast verification practice [11], [12].

For a continuous binary score $s(x)$ and decision threshold τ , the predicted class is

$$\hat{y} = \mathbb{I}[s(x) \geq \tau]. \quad (24)$$

The receiver-operating-characteristic quantities are

$$\text{TPR}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \quad \text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)}. \quad (25)$$

ROC-AUC is the area under the receiver-operating-characteristic curve,

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(u) du. \quad (26)$$

For continuous scores, it can also be interpreted as

$$\text{AUC} = P(s(x^+) > s(x^-)), \quad (27)$$

up to the usual tie convention. Here x^+ and x^- represent randomly selected positive and negative examples.

The precision for a threshold τ is

$$\text{Precision}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FP}(\tau)}. \quad (28)$$

PR-AUC is the area under the precision-recall curve. Because M- and X-class events are rare, PR-AUC is particularly useful for understanding positive-class behavior alongside ROC-AUC.

C. Reproducibility environment

The experiments are associated with the repository environment and the following command-line entry points:

```
venv/bin/python -m src.train_xgboost --target ge_c
venv/bin/python -m src.train_xgboost --target ge_m
venv/bin/python train_histgb.py --target ge_c
venv/bin/python train_histgb.py --target ge_m
venv/bin/python train_extratrees.py --target ge_c
venv/bin/python train_extratrees.py --target ge_m
```

The repository stores XGBoost outputs under `experiments/xgboost/`, HistGradientBoosting outputs under `experiments/histgb/`, and Extra Trees outputs under `experiments/extratrees/`. CNN-LSTM artifacts include checkpoints, a classification report, training-history CSV, plots, and confusion matrices.

The documented software environment reports a ROCm-enabled PyTorch build, but the execution environment exposed no usable GPU device (`torch.cuda.is_available()` returned false and `/dev/kfd` was unavailable). Accordingly, the reported experiments should be interpreted as CPU runs unless a separate artifact explicitly states otherwise.

TABLE IV
CNN-LSTM TEST PERFORMANCE ON THE REPORTED C/M/X
MULTICLASS TASK.

Metric	Test value
Accuracy	0.825829
Macro recall / POD	0.457667
Macro F1	0.444940
Weighted F1	0.825060
Macro FAR	0.231867
Macro CSI	0.367100
Macro TSS	0.261891
Macro HSS	0.248024

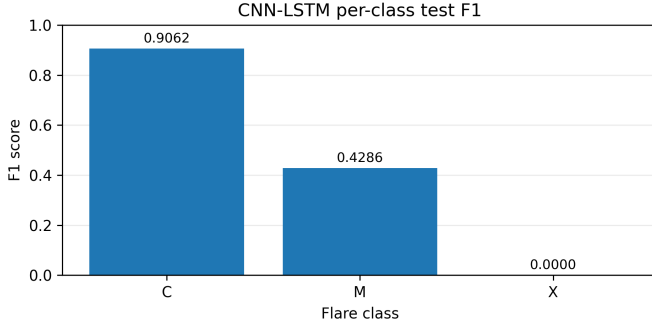


Fig. 3. CNN-LSTM F1 score by flare class on the held-out test set. The X class has zero F1 in the reported run.

VI. RESULTS

A. CNN-LSTM multiclass results

The CNN-LSTM achieves an overall test accuracy of 0.825829 and a macro F1 of 0.444940. Its weighted F1 is 0.825060, while macro recall/POD is 0.457667. The corresponding macro FAR, CSI, TSS, and HSS are 0.231867, 0.367100, 0.261891, and 0.248024, respectively.

The per-class F1 scores are 0.9062 for C, 0.4286 for M, and 0.0000 for X. The X class contains only 34 examples in the test set and was not detected by this run. This is the most important caveat in the multiclass result: the headline accuracy is not a sufficient summary of performance on rare, high-impact events.

B. Binary tabular results

Table V summarizes the reported held-out test results. A dash denotes a metric that was not present in the supplied experiment record rather than a value of zero.

C. Model-wise ranking performance

The aggregate ROC-AUC values show that XGBoost is the strongest ranking model for both target definitions, while

logistic regression remains competitive, particularly for the $\geq M$ target. Figure 4 displays the reported values as model-wise trajectories rather than a conventional bar chart.

For the $\geq C$ target, XGBoost provides the highest ROC-AUC (0.890780) and PR-AUC (0.852837). Its TSS of 0.502974 is nevertheless slightly below logistic regression (0.537476) and HistGradientBoosting (0.520046). Thus, ranking quality and thresholded forecast skill are not equivalent properties.

D. Model-wise thresholded skill

Figure 5 shows the same model-wise structure for TSS. Unlike ROC-AUC, TSS is tied to a threshold and explicitly balances hit rate against false-positive rate.

For the $\geq M$ target, XGBoost is strongest on the reported ROC-AUC (0.939496), PR-AUC (0.720826), and TSS (0.757143). Its recall is 1.000000, but precision is only 0.333333. Thus, all positive cases are detected at the selected threshold, but a substantial proportion of positive predictions are false alarms.

E. ROC-space operating points

The complete prediction-score vectors required to reconstruct full ROC curves were not preserved in the supplied experiment record. Consequently, a full ROC curve would require inventing intermediate points. Instead, the thresholded ROC-space points below are derived exactly from the reported recall and TSS values.

From

$$\text{TSS} = \text{TPR} - \text{FPR}, \quad (29)$$

the false-positive rate at a reported operating point is

$$\text{FPR} = \text{TPR} - \text{TSS}. \quad (30)$$

The resulting values are

$$\text{FPR}_{XGB, \geq C} = 0.941176 - 0.502974 = 0.438202, \quad (31)$$

$$\text{FPR}_{HGB, \geq C} = 0.763903 - 0.520046 = 0.243857, \quad (32)$$

$$\text{FPR}_{ET, \geq C} = 0.711447 - 0.505629 = 0.205818, \quad (33)$$

$$\text{FPR}_{XGB, \geq M} = 1.000000 - 0.757143 = 0.242857, \quad (34)$$

$$\text{FPR}_{HGB, \geq M} = 0.583031 - 0.537638 = 0.045393, \quad (35)$$

$$\text{FPR}_{ET, \geq M} = 0.701691 - 0.605207 = 0.096484. \quad (36)$$

The ROC-space view makes the threshold trade-off more intuitive than a bar chart. XGBoost reaches the highest TPR

TABLE V
TABULAR-MODEL TEST PERFORMANCE FOR THE TWO BINARY FORECASTING TARGETS. THRESHOLDS ARE SHOWN WHERE EXPLICITLY REPORTED.

Target	Model	Thr.	ROC-AUC	PR-AUC	TSS	F1	Precision	Recall
$\geq C$	XGBoost	0.2	0.890780	0.852837	0.502974	0.748538	0.621359	0.941176
$\geq C$	Logistic Regression	–	0.846086	–	0.537476	–	–	–
$\geq C$	HistGradientBoosting	0.2	0.833828	0.647268	0.520046	0.589272	0.479627	0.763903
$\geq C$	Extra Trees	0.3	0.830317	0.632153	0.505629	0.590177	0.504228	0.711447
$\geq M$	XGBoost	0.1	0.939496	0.720826	0.757143	0.500000	0.333333	1.000000
$\geq M$	Logistic Regression	–	0.922494	–	0.581815	–	–	–
$\geq M$	HistGradientBoosting	0.1	0.891795	0.381298	0.537638	0.411113	0.317494	0.583031
$\geq M$	Extra Trees	0.1	0.901546	0.313543	0.605207	0.321461	0.208487	0.701691

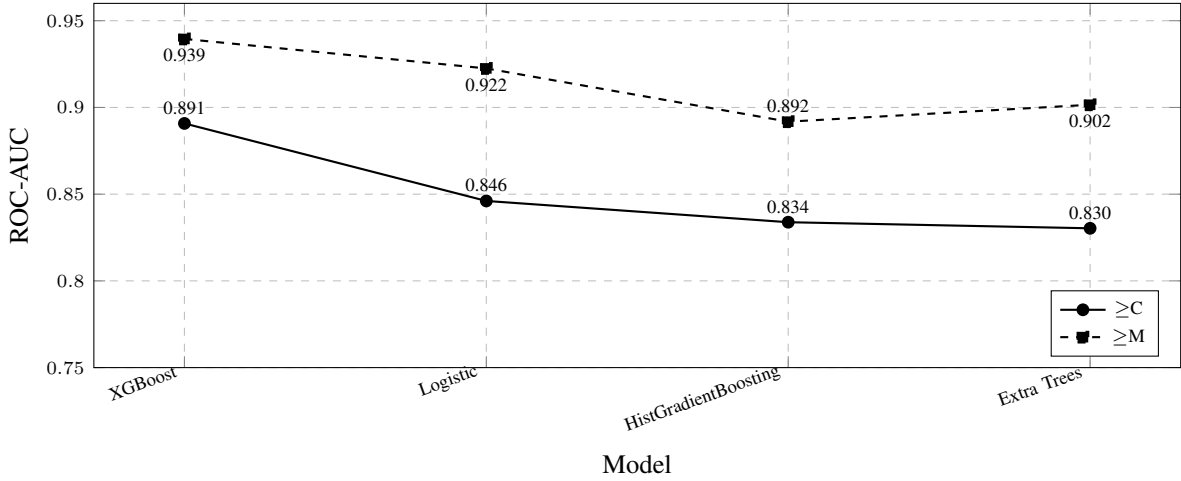


Fig. 4. Model-wise ROC-AUC trajectories for the two binary targets. Each point is an experimentally reported test value; connecting lines only organize the model comparison and do not represent a continuous ROC curve.

for $\geq M$, but does so with an FPR of approximately 0.243. HistGradientBoosting and Extra Trees use lower FPR operating points for $\geq M$, with lower recall.

F. Precision–recall operating points

The precision–recall plane provides a complementary view because it emphasizes the quality of positive predictions. This is particularly relevant under severe class imbalance.

The XGBoost trajectory shows a particularly clear trade-off: recall rises from 0.941176 for $\geq C$ to 1.000000 for $\geq M$, while precision falls from 0.621359 to 0.333333.

VII. DISCUSSION

A. Complementary, not interchangeable, experiments

The principal methodological point is that the five models are not solving a single identical classification problem. The CNN-LSTM predicts among C, M, and X using image sequences, while the four tabular classifiers predict two binary

thresholds using 29 engineered scalar features. Consequently, the 0.825829 CNN-LSTM accuracy should not be compared numerically with a binary tabular accuracy and interpreted as evidence that one representation is universally better.

Instead, the results answer two related questions: how well a short sequence of magnetograms supports multiclass flare identification, and how well engineered magnetic-complexity descriptors support binary event detection.

B. Class imbalance and rare-event behavior

The CNN-LSTM results make the class-imbalance issue visible: C-class F1 is 0.9062, M-class F1 is 0.4286, and X-class F1 is 0.0000. The weighted F1 nearly matches the overall accuracy because the more frequent class dominates the weighted aggregate. Macro metrics are therefore the more informative summary for the multiclass experiment.

The binary results show a complementary failure mode. For $\geq M$, XGBoost detects every positive example at the selected

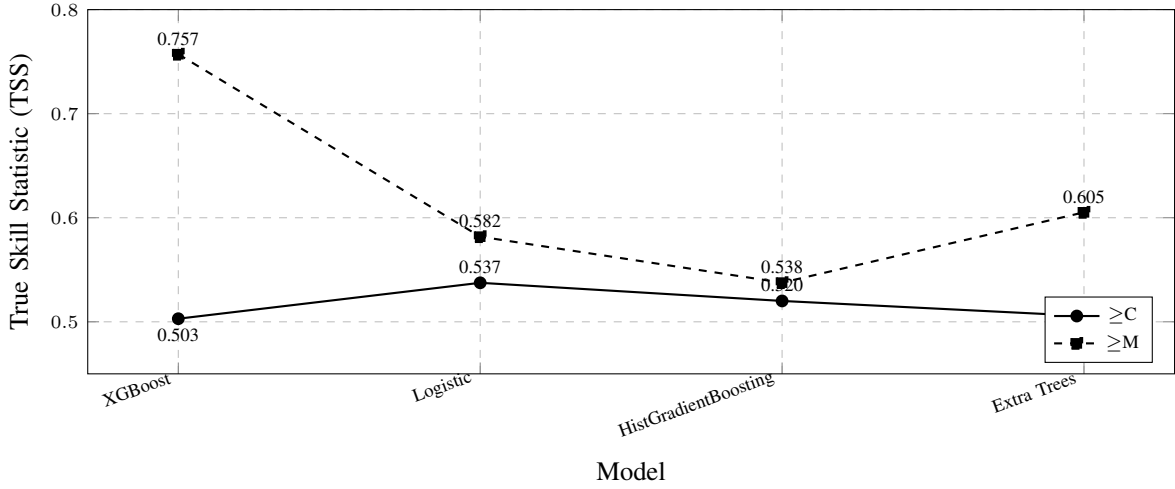


Fig. 5. Model-wise TSS trajectories for the two binary targets at the reported operating points. The higher $\geq M$ TSS values show the stronger discrimination achieved for the more severe flare threshold by several models.

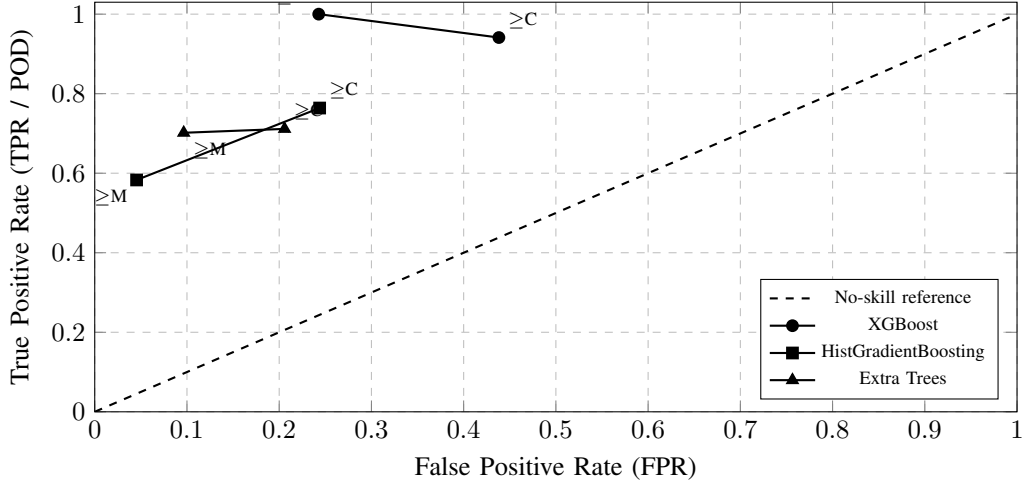


Fig. 6. ROC-space operating points for the tree-based models. Each line connects the measured $\geq C$ and $\geq M$ operating points of one model. The segments are visual guides, not reconstructed ROC curves.

threshold but also generates many false alarms. A high recall is operationally meaningful only if the corresponding false-alarm burden is acceptable. This aligns with established flare-forecast verification practice in which TSS, FAR, CSI, and HSS are reported alongside conventional classification metrics [11].

C. Ranking performance versus operating-point skill

The results demonstrate why ROC-AUC should not be treated as synonymous with forecast accuracy. ROC-AUC measures the ability to rank positive cases above negative cases across possible thresholds, whereas TSS evaluates a specific operating point. For $\geq C$, XGBoost has the strongest reported ROC-AUC and PR-AUC, but logistic regression has

slightly higher TSS at its reported operating point. This is not contradictory: the two metrics answer different questions.

The distinction is even clearer for $\geq M$. XGBoost has ROC-AUC 0.939496 and TSS 0.757143, but the selected threshold produces precision of only 0.333333. The model is therefore highly discriminative in ranking terms while still being costly if false alarms are operationally expensive.

D. Relation to prior work

The tabular performance is broadly consistent with the established finding that magnetic-complexity descriptors carry useful predictive information. The TSS of 0.757143 for $\geq M$ is within the range of strong results reported in previous studies, although direct comparison is not scientifically valid

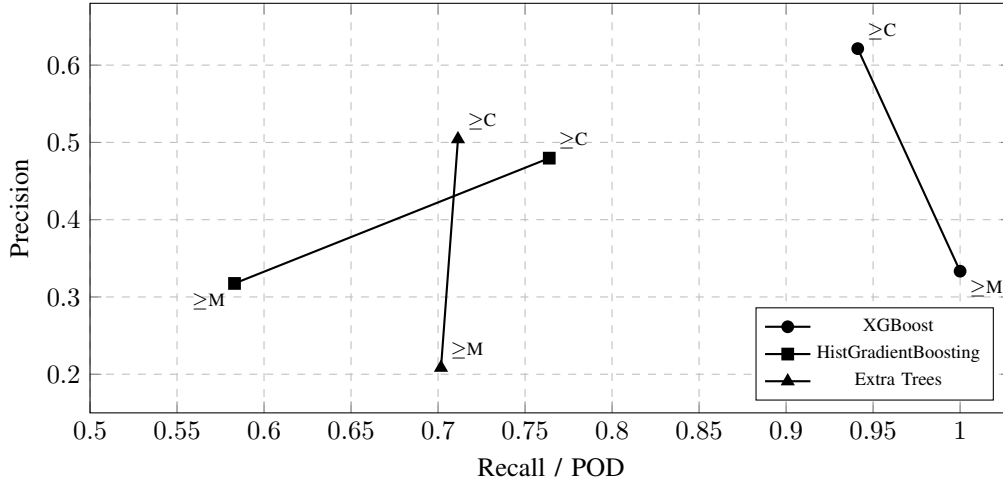


Fig. 7. Precision–recall operating points at the reported test thresholds. The movement toward the $\geq M$ target reveals the increase in false-alarm burden, especially for XGBoost. Connecting segments are visual guides between measured operating points and are not reconstructed PR curves.

without matching labels, forecast windows, data partitions, and threshold-selection procedures. Nishizuka *et al.* reported strong TSS values for $\geq M$ and $\geq C$ flare forecasting using a different feature set, time period, and operational split [8]. Florios *et al.* likewise reported strong TSS with a different SDO/SHARP predictor design [6]. These studies are best viewed as contextual reference points, not as direct baselines.

The CNN-LSTM branch is aligned with a growing body of work showing that temporal evolution can be informative. Sun *et al.* found benefits from LSTM-based representations across solar cycles, while Chen *et al.* demonstrated that spatial and temporal representations of HMI observations can be used to identify flare precursors [10], [13]. The present result does not establish that a CNN-LSTM is superior to a purely spatial model because a matched single-frame control result is not part of the finalized experiment record.

Recent transformer-based studies further motivate controlled temporal representation comparisons. Alatoom and Nikolaou reported an active-region transformer framework for 24-hour forecasting, while Riggi *et al.* evaluated foundational transformer models across image, video, and time-series modalities [21], [22]. These results suggest useful future baselines, but should be evaluated under the same grouped and chronological protocols to preserve scientific comparability.

E. Why the 29-feature tabular branch remains valuable

The engineered features provide a low-dimensional, interpretable representation of magnetic complexity. This makes them attractive for sensitivity studies, threshold selection, model diagnostics, and future hybrid models. The feature

groups are physically motivated: gradients characterize spatial field variation, neutral-line descriptors summarize polarity-interface geometry, wavelets represent multiscale magnetic structure, and flux quantities describe the overall magnetic content of the region [4], [19].

A natural next step is therefore not to replace the tabular branch with the image branch, but to investigate whether learned CNN representations and physically interpretable features provide complementary information. This is consistent with the broader gray-box direction proposed for space-weather machine learning [12].

F. Overfitting and temporal validity

The active-region-grouped partition reduces a major source of data leakage because the same active region does not occur in both training and evaluation. However, there is not enough saved information to conclude that all overfitting has been eliminated. Systematic train-versus-validation learning curves are not available for every model, and the predefined timestamp ranges overlap. The results should therefore be interpreted as grouped benchmark performance rather than proof of chronological generalization.

VIII. LIMITATIONS AND FUTURE WORK

This benchmark has five principal limitations. First, the CNN-LSTM and tabular models use different task formulations, so the study does not provide a single matched head-to-head comparison. Second, the X-class sample in the CNN-LSTM test set is extremely small and resulted in zero detected X-class examples in the reported run. Third, binary class

imbalance creates a trade-off between detection and false alarms that cannot be summarized by recall alone. Fourth, the experiment uses active-region-grouped but not chronological partitions, so performance on a truly future solar interval remains untested. Fifth, the repository does not contain a complete train/validation learning-curve analysis for every model, limiting strong conclusions about generalization.

Future work should address these limitations by constructing a chronological out-of-time test set, repeating the experiment over multiple active-region partitions, calibrating predicted probabilities, and reporting confidence intervals across repeated runs. The available SHARP/NASA-related CSV can then be used for an explicitly external-validation study after feature alignment and label reconciliation. A matched image-versus-feature experiment should also evaluate the same binary target on both representations so that the value of temporal imaging can be measured directly.

From the modeling perspective, future experiments can test focal or cost-sensitive objectives for the rare M/X classes, longer sequences, alternative temporal encoders, and hybrid models that concatenate learned image representations with physically engineered features. Recent transformer-based studies suggest that attention mechanisms and pretrained visual representations may provide another useful comparison point, but such models should be evaluated under the same grouped and chronological protocols to preserve scientific comparability [21], [22].

IX. CONCLUSION

This paper presented a reproducible benchmark of spatiotemporal and tabular machine-learning methods for solar-flare prediction using SDO/HMI active-region observations. The reported CNN-LSTM achieved 0.825829 test accuracy and 0.444940 macro F1 on the C/M/X multiclass problem, but its zero X-class F1 demonstrates that high overall accuracy can conceal failure on rare, high-impact events.

Across the binary threshold experiments, XGBoost was the strongest overall tabular model, reaching ROC-AUC values of 0.890780 and 0.939496 for $\geq C$ and $\geq M$, respectively, and a TSS of 0.757143 for $\geq M$. That $\geq M$ result nevertheless combines perfect recall with only 0.333333 precision, underscoring the importance of false-alarm-aware evaluation.

The main conclusion is therefore methodological rather than a claim of universal model superiority: solar-flare prediction should be evaluated with task-aware metrics, explicit leakage

controls, and separate reporting of common and rare classes. The current results provide a useful baseline for a next stage in which image-derived temporal representations, engineered magnetic features, and external SHARP information are evaluated under matched binary and chronological forecasting protocols.

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